

C0 – 2 – ASSESSMENT TOOL 3 – PROBLEM STATEMENT

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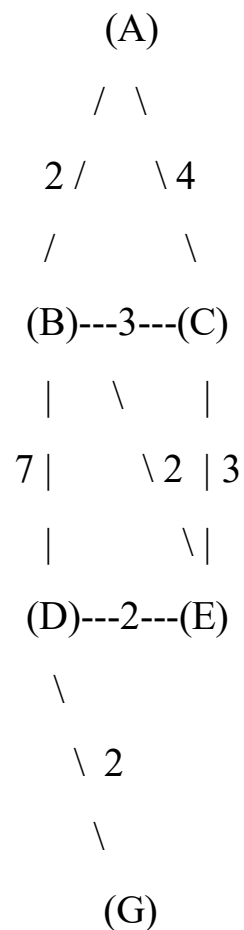
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COURSE NAME : ARTIFICIAL INTELLIGENCE

Question 1: Dry Run of A Search Algorithm

Consider the following graph.



Heuristic Values

Node	$h(n)$
A	7
B	6
C	4
D	3
E	2
G	0

Task

Perform a dry run of the A* Search algorithm to find the shortest path from A to G.

For each iteration, show the following:

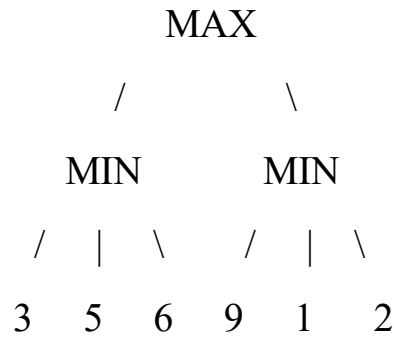
- Current node
- Open List
- Closed List
- $g(n)$
- $h(n)$
- $f(n) = g(n) + h(n)$

Finally

1. Identify the optimal path.
2. Calculate the total path cost.

Question 2: Dry Run of Minimax with Alpha-Beta Pruning

Consider the following game tree.



Task

Perform a dry run of the Minimax algorithm with Alpha-Beta Pruning by evaluating the tree from left to right.

For each node, show:

- Alpha (α)
- Beta (β)
- Selected value
- Any node(s) that get pruned

Finally

1. Determine the best move for MAX.
2. State the final minimax value.
3. List the pruned node(s) (if any).